

Gibbs free energy and standard electrode potentials (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R3.2.13) Predict whether a reaction is spontaneous in the forward or reverse direction from $E^\ominus$ data.
- (R3.2.14) Determine the value for $\Delta G^\ominus$ from $E^\ominus$ data.

Calculations involving standard electrode potentials

Now that you have a better understanding of how we determine the standard electrode potential for a certain half-cell reaction and the placement of a species within the reactivity series, you can take this information even further and extend it to more complex voltaic and electrolytic cells.

Take for example primary and secondary batteries (see [section R3.2.6–7](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/voltaic-cells-and-rechargeable-batteries-id-46490/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/voltaic-cells-and-rechargeable-batteries-id-46490/>)). These types of batteries, often referred to simply as disposable and rechargeable, must provide a consistent output of electricity to operate a device. The amount of electrical charge generated from a fresh disposable battery differs from the energy produced by a depleted rechargeable battery. But why are some batteries rechargeable, and others not? What determines the voltage produced by a particular battery, and how does the chemical composition of the battery affect this value?

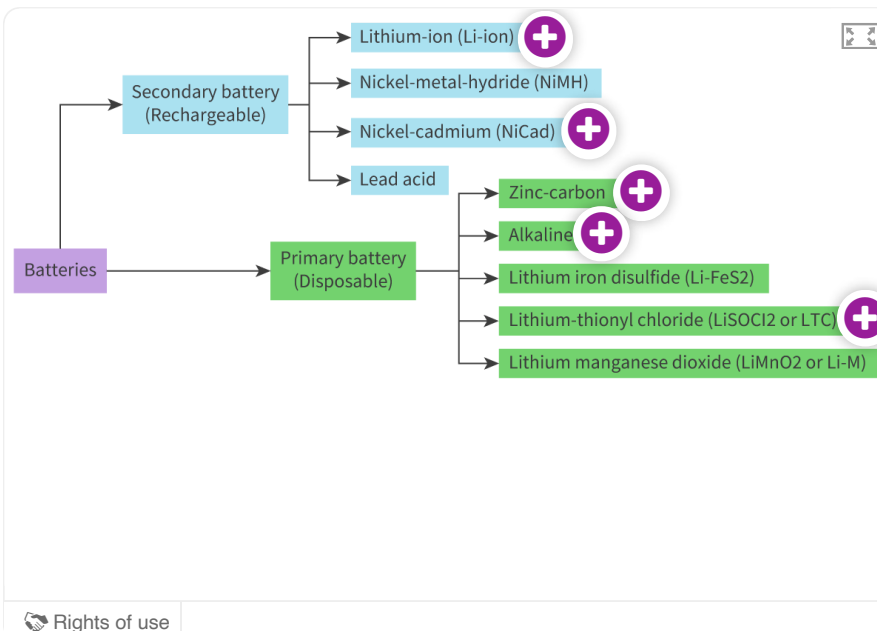
International Mindedness

Lithium ion batteries are extraordinarily useful in electrically powered vehicles (EVs) as lithium is the most easily oxidised species of metal.

EVs are essential to modernising the automotive industry as the global community moves to phase out classic combustion engine vehicles in order to reduce greenhouse gas emissions.

But where does this lithium come from and what impact does the mining of this metal have on the global ecosystem? Can the global demand for lithium be met as we move towards having only EVs on our roads?

Some examples are given in the flow chart in **Interactive 1**.



Interactive 1. Examples of Primary and Secondary Batteries.

More information for interactive 1

An interactive flowchart illustrates two major types of batteries: primary (disposable) and secondary (rechargeable), along with several examples for each category. Some examples feature a plus sign, representing an interactive hotspot. Selecting the hotspot reveals photographs of the respective battery types.

Secondary Batteries (Rechargeable): These batteries are rechargeable and can be used multiple times. A reversible chemical reaction allows electrons to be restored to the battery through charging.

Examples of secondary batteries shown in the flowchart are as follows:

1. Lithium-ion (Li-ion) with a hotspot: Photo of a rectangular lithium-ion battery labeled lithium-ion battery 3.7 V, 3000 mAh. The battery also shows a recycling symbol with a crossed-out trash can, which means it should not be thrown in the regular trash can.
2. Nickel-metal hydride (NiMH)
3. Nickel-cadmium (NiCad) with a hotspot: Photo of a cylindrical rechargeable battery labeled Nickel cadmium made by Berec. The full text on the battery reads: Rechargeable sealed cell; 1.2v, 2.0Ah; charge at 250 mA for 12 hours or as advised; NCC 200 C Size; KAH 27/50.
4. Lead acid

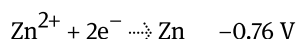
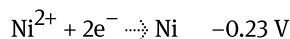
Primary Batteries (Disposable): These batteries are designed for one-time use. Once the chemicals are spent, they cannot be recharged.

Examples of primary batteries shown in the flowchart are as follows:

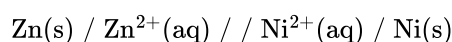
1. Zinc-carbon with a hotspot: Photo of an AA-size cylindrical battery labeled KODAK Super Heavy Duty ZINC AA R6P 1.5V.
2. Alkaline with a hotspot: Photo of a cylindrical Kodak Xtralife AAA alkaline battery labeled, Alkaline Battery AAA LR03 1.5V.
3. Lithium iron disulphide (Li-FeS₂)
4. Lithium thionyl chloride (LiSOCl₂ or LTC) with a hotspot: Photo of a rectangular Sony Lithium Thionyl Chloride Battery with wires attached.
5. Lithium manganese dioxide (LiMnO₂ or Li-M)

This flowchart supports the understanding of electrochemical cells by showing real-world applications of electrode potentials and redox reactions. By analyzing the chemical compositions and structures of these batteries, viewers can relate the theoretical concepts of voltage and energy capacity to actual battery performance.

If you take the standard electrode potentials for nickel and zinc it is clear that if the two half-reactions were to be combined to form a voltaic cell, then zinc would be preferentially oxidised and nickel would be reduced:



Recall the concept of electrochemical cell notation (cell diagrams) (see [section R3.2.3–4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ease-of-oxidation-and-reduction-id-46488/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ease-of-oxidation-and-reduction-id-46488/>)). In this example, we would write the cell diagram as:



since the zinc is being oxidised (at the anode) and nickel is being reduced (at the cathode).

It is possible to then determine the overall standard cell potential for this reaction, by subtracting the standard electrode potential of the anode (oxidised species) from the standard electrode potential for the cathode (reduced species).

$$E_{\text{cell}}^{\ominus} = E^{\ominus}(\text{cathode or reduced species}) - E^{\ominus}(\text{anode or oxidised species})$$

For this particular example involving zinc and nickel, the calculation would be.

$$\begin{aligned} E_{\text{cell}}^{\ominus} &= E^{\ominus}(\text{nickel}) - E^{\ominus}(\text{zinc}) \\ &= -0.23 - (-0.76) \\ &= +0.53 \text{ V} \end{aligned}$$

The standard cell potentials do not need to be multiplied by any coefficient even if coefficients are needed to balance the chemical reaction. As well, by convention the standard cell potentials are always given as reduction potentials and so the values are never inverted within the calculation even if a species is undergoing oxidation.

Positive standard cell potentials are associated with spontaneous reactions, whereas negative cell potentials are associated with non-spontaneous reactions. Non-spontaneous electrochemical reactions include such processes as the electrolysis of water and electroplating, which you will study in more detail in [sections R3.2.15](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrolysis-of-water-and-aqueous-solutions-hl-id-46495/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrolysis-of-water-and-aqueous-solutions-hl-id-46495/>) and [R3.2.16](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electroplating-hl-id-46496/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electroplating-hl-id-46496/>).

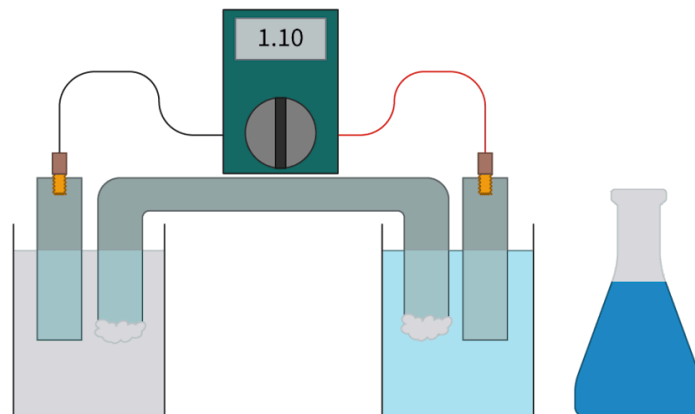


Figure 1. The experimental setup for a voltaic cell involving copper and zinc electrodes in solutions of copper sulfate and zinc sulfate.

 More information for figure 1

The image shows an experimental setup for a voltaic cell. On the left, a beaker contains a copper electrode immersed in a solution, likely copper sulfate, indicated by a light blue color. On the right, another beaker holds a zinc electrode in a light blue solution, likely zinc sulfate. These two beakers are connected by a U-shaped tube, suggesting a salt bridge, allowing ions to flow between the two solutions to maintain the charge balance. A voltmeter, placed at the center top, displays a reading of 1.10 volts, measuring the potential difference between the two electrodes. Wires connect the electrodes to the voltmeter, with the left side having a black wire and the right side a red wire, indicating polarity. A partially filled blue laboratory flask is placed next to the zinc beaker, indicating the presence of a chemical reagent possibly involved in the reaction.

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Worked example 1

The redox reaction between zinc and copper in solutions of zinc sulfate and copper(II) sulfate produces a spontaneous electrochemical reaction.

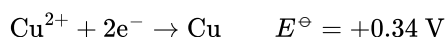
Using the standard electrode potentials given in [section 19](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/standard-reduction-potentials-at-29815-k-id-45121/>) of the DP chemistry data booklet, justify why this reaction is classified as spontaneous based on its overall standard cell potential.

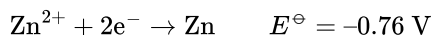
$$\begin{aligned} E_{\text{cell}}^{\ominus} &= E^{\ominus} \text{ Cu (reduced species)} - E^{\ominus} \text{ Zn (oxidised species)} \\ &= +0.34 - (-0.76) \\ &= +1.10 \text{ V} \end{aligned}$$

As the $E_{\text{cell}}^{\ominus}$ is positive, the overall voltaic reaction is defined as being spontaneous.

In order to complete this question you must understand that since copper is placed in a solution of copper(II) sulfate the half-cell reaction for copper is:



The half-reaction for zinc is:



Therefore, zinc will preferentially be oxidised over copper.

The anode will be zinc and the cathode will be copper.

Gibbs free energy

It is interesting to realise that many electrochemical reactions are reversible processes. During the day when you are using your cell phone, its battery is acting as a spontaneous voltaic cell, where the flow of electrons from the anode to the cathode is powering your device. At night, when you charge your phone, the electrical current from the wall outlet supplies the flow of electrons that allows for the chemical reaction inside your battery to proceed in the reverse (and non-spontaneous) direction. When you wake up in the morning the battery is fully 'charged' and the voltaic cell is ready to be used once again. That is of course if you remember to charge it.

How does this relate to Gibbs free energy? How is it possible to determine the from $\Delta G^{\ominus}$ from the standard cell potential of an electrochemical reaction?



Figure 2. Charging a battery favours the non-spontaneous reaction.

Credit: Priscila Zambotto, Getty Images

This is not the first time that you have studied the idea of spontaneous versus non-spontaneous chemical reactions. You may recall that changes in Gibbs free energy allow you to determine if a reaction is spontaneous or not in a particular direction (see [section R1.4.2-3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/>)). If the change in Gibbs free energy ($\Delta G^{\ominus}$) for a reaction is negative, then the reaction is said to be spontaneous in the forward direction under standard conditions.

Nature of Science

Aspect: Theories

It is interesting to note that the concept of Gibbs free energy is present in many areas of chemistry, despite being originally developed by Josiah Willard Gibbs as a theory related to thermodynamics. From biochemistry, to thermodynamics, to equilibria, to electrochemical cells, Gibbs free energy allows us to tie together concepts like enthalpy, entropy, equilibrium position, rates of reaction, and electromotive forces under one unifying umbrella. The greatest discoveries made in science relate to concepts and theories that transcend the field in which they were developed and help to create unifying fundamental principles within all of the sciences.

It is possible to determine the change in Gibbs free energy from the standard cell potential of a particular electrochemical reaction by using the equation:

Table 1. Equation for change in Gibbs free energy.

Equation	Symbols	Units
$\Delta G^\ominus = -nFE^\ominus$	$\Delta G^\ominus$ is the change in standard Gibbs free energy	joules per (J mol ⁻¹)
	n is the number of moles of electrons transferred	
	F is the Faraday constant ($9.65 \times 10^4 \text{ C mol}^{-1}$)	Coulombs per (C mol ⁻¹)
	$E^\ominus$ is the standard cell potential	Volts (V)

From this equation it is possible to observe that the change in Gibbs free energy and standard cell potential will always have values that are opposite to one another for a given reaction. If a reaction is spontaneous in the forward direction, then the standard cell potential will be positive and the associated change in Gibbs free energy will be negative, and vice versa for the reverse reaction which by default would be non-spontaneous. Recall that as a reaction approaches equilibrium the change in Gibbs free energy starts to approach zero (see [section R1.4.4a \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-equilibrium-hl-id-44767/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-equilibrium-hl-id-44767/)). This is true for both thermodynamic and electrochemical systems.

Study skills

It can be difficult to recall all the appropriate units and constants used for various types of calculations.

Note that the Faraday constant can be found in the DP chemistry data booklet.

When performing this particular calculation using

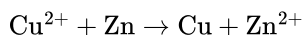
$$\Delta G^\ominus = -nFE^\ominus$$

it is important to understand that a potential difference of one volt is equivalent to one joule per coulomb.

$$1 \text{ volt} = 1 \text{ J C}^{-1}$$

Worked example 2

In the worked example from earlier in this section it was discovered that the voltaic reaction between copper and zinc produces a standard cell potential ($E_{\text{cell}}^{\ominus}$) of +1.10 V for the reaction:



Calculate the value of $\Delta G^{\ominus}$ for this reaction.

Note that two electrons are being transferred from the zinc to the copper ions, therefore $n = 2$.

$$1 \text{ V} = 1 \text{ J C}^{-1}$$

$$\begin{aligned}\Delta G^{\ominus} &= -nFE^{\ominus} \\ &= -2 \times 9.64 \times 10^4 \times 1.10 \\ &= -2.12 \times 10^5 \text{ J mol}^{-1} \text{ or } -212 \text{ kJ mol}^{-1}\end{aligned}$$

Now you will be able to take these concepts and apply them to other situations.

Activity

- **IB learner profile attribute:**
 - Inquirer
 - Reflective
- **Approaches to learning:**
 - Thinking skills — Reflecting on the credibility of results
 - Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 20–30 minutes
- **Activity type:** Individual/pair activity

Using the simulator, build four different voltaic cells with a variety of combinations of electrodes and electrolytes (all at a concentration of 1.0 M), and compare your experimental values to the theoretical standard cell potentials which you will calculate yourself using the standard cell potentials provided in the DP chemistry data booklet.

1. Access the simulator via the link: <https://teachchemistry.org/classroom-resources/galvanic-voltaic-cells-2> (<https://teachchemistry.org/classroom-resources/galvanic-voltaic-cells-2>)
2. Under the heading Galvanic/Voltaic Cells 2 click on Start →
3. Create a data table in Excel or using a digital notebook.
 - (a) Create a table with three columns and five rows. The first column should have the heading 'Cell Combinations' for the reactions between the species, zinc and silver, zinc and magnesium, copper and silver, and lastly silver and magnesium.
 - (b) The second column should have the heading 'Standard Cell

Potential' and a third column with 'Cell Notation'

4. Using the simulator, construct a voltaic cell consisting of a zinc electrode in a 1.00 M zinc nitrate solution in the left beaker and a silver electrode in 1.00 M $\text{AgNO}_3(\text{aq})$ in the right beaker.
5. Start the simulation by clicking the button on the voltmeter from off to on.
6. Record any quantitative data (including measurement uncertainties) for the standard cell potential.
7. Repeat the experiment three more times using the other combinations of electrodes and solutions that you recorded in your data table.

Use the data recorded during the activity to answer these questions.

1. Using the standard electrode potentials in the DP chemistry data booklet, calculate the theoretical standard cell potentials for the four simulated reactions that you ran.
2. Complete the data table by inserting the correct cell notation for each of the four reactions that were performed in this investigation. Be careful to understand which species is being oxidised and which is being reduced.
3. Explain why it is necessary to keep the concentrations of the aqueous solutions at 1 mol dm^{-3} .
4. List the reactions in order from most spontaneous to least spontaneous, using calculations of the change in Gibbs free energy to justify why they are listed in this order.
5. Deduce how the results of your investigation may have been different in terms of accuracy and precision if you had conducted this experiment using a hands-on experimental setup instead of a simulation.